UNDER REVIEW

RESCIENCE C

Replication / Natural Language Processing

Does positional bias in LLM-as-judge evaluation, and its calibration fix, hold for open-weight judges? A pre-registered replication halted by its own determinism control

immediate

October 13, 2026

Edited by
(Editor)

Received
–

Published
–

DOI
–

DRAFT, NOT A SUBMISSION. Pre-registration: https://osf.io/jw59c/ (DOI 10.17605/OSF.IO/JW59C).

Abstract

Wang et al. [1] report that large language models used as pairwise evaluators reverse their verdicts when the two candidate answers are swapped, and propose three calibration strategies, one of which this study replicates, Multiple Evidence Calibration, that recover alignment with human labels. Their judges were closed OpenAI endpoints, and the snapshot pinned in their repository is not in the provider's current model list, so their numbers can no longer be reproduced with the original judges. We pre-registered a replication that repeats their published protocol while substituting three open-weight instruction-tuned judges, and report it here.

None of the three pre-registered hypotheses is evaluable as registered. All three read from a temperature-0 arm that did not survive: a determinism control we registered before running found that two of the three judges failed to reproduce byte-identical output at temperature 0 with a fixed seed, which halted their temperature-0 arms by the rule we had committed to, and the third judge's output did not satisfy the registered output-format parser for 313 of its 320 records. H3 was additionally unevaluable for a reason independent of that: one of its two required cells was never generated, and no counterpart to the original's calibrated quantity was ever computed. H2 was not evaluable as registered, because the registered Table 1 temperature-0 VANILLA arm produced no analysable data. Under a post-registration deviation (the deviations register, D-8), fixed before any H2 statistic was computed, the unchanged test was applied to every surviving both-order MEC arm and the descriptive criterion was met on SCORING_CODE - one pair, Llama-3.1 against Qwen2.5, with opposite signs and non-overlapping intervals - and was not met on COMPARING. H2 remains NOT evaluable as registered; these results are descriptive only. Under the asymmetry declared in advance, a null is never written as a refutation. A single pre-declared re-test under serial requests, run on the one judge that had failed first (Qwen2.5-7B-Instruct), found the non-determinism persisted for that judge under the one controlled condition we tested, serial requests to a server we intended to be otherwise idle with the model resident, conditions the re-test script records but does not verify, which is narrower than "contention removed" in general, so we do not attribute it to load.

Copyright © 2026 V. K. Sunkavalli, released under a Creative Commons Attribution 4.0 International license.
Correspondence should be addressed to Veerendra Kumar Sunkavalli (veeru.svk@gmail.com)
The authors have declared that no competing interests exist.
Code is available at https://github.com/dosvk/faireval-openweight-replication.

The temperature-1 arms survived and are reported descriptively. On those arms, and as a description rather than a test, verdicts changed with presentation order beyond each judge's own same-order noise - by 0.038, 0.138 and 0.250 on the scoring template, and most strongly by 0.810 on the reconstructed COMPARING prompt. We also report what the failures cost, the parse accounting per judge and arm, one operational data loss of our own, and every deviation from the registered plan, each dated. The replication's contribution is therefore not a verdict on positional bias. It is a demonstration that a protocol resting on temperature-0 determinism and on a strict output format can fail to execute on open-weight judges, and a record of exactly where and how it failed.
Pre-registration: https://osf.io/jw59c/ (DOI 10.17605/OSF.IO/JW59C).

1 Introduction

Large language models are now routinely used to score other models' outputs. The appeal is obvious: a judge model is cheaper than a human annotator and available on demand. The risk is equally obvious, and Wang et al. [1] demonstrated one form of it. They showed that an LLM asked to pick the better of two answers can reverse its verdict when the two answers are swapped in the prompt, reported conflict rates as high as 82.5% under such a swap, and proposed three calibration strategies. This study replicates ONE of them, Multiple Evidence Calibration. Balanced Position Calibration was not run: no BPC quantity was computed at any point, which is one of two independent reasons H3 is not evaluable (see the deviations register).

That result matters to anyone who uses a judge model, and it invites a direct question. The original used two closed OpenAI endpoints as judges. The snapshot pinned in its repository, `gpt-3.5-turbo-0301`, is not in the provider's current model list, so the original's numbers can no longer be reproduced with the original judges by anyone, including its authors. Does the finding survive when the judge is an open-weight model that anyone can download and pin by digest?

We pre-registered a replication to answer that.[2] Registration preceded every confirmatory arm. It did not precede all contact with the data: the registration itself discloses a 6-call pilot and a 9-call output-format check on 3 of the 80 questions, run before registering (https://osf.io/jw59c/, DOI 10.17605/OSF.IO/JW59C). The design repeats the published protocol and varies one technical detail deliberately, the judge, which is the kind of variation a replication involves. Three open-weight instruction-tuned models from three different families stand in for those endpoints.

The answer is that we could not run the registered test, and the reason is itself the finding. The pre-registration included a determinism control, written before any run, requiring that re-running a prompt at temperature 0 with a fixed seed reproduce byte-identical output. The original states that it sets "the generated temperature to 0 for deterministic generation results", and H1 and H3 both read from that temperature-0 arm, and H2's scope is discussed in the deviations register. Two of our three judges failed the control. The registered rule halted their temperature-0 arms, and we did not convert the failure into a correction term, because the pre-registration explicitly forbids that. The third judge passed the control but produced output that did not satisfy the registered output-format parser on 313 of 320 records, leaving one usable paired observation.

So H1 and H3 are not evaluable as registered, and H2, which the registration binds to no particular arm, is also not evaluable as registered; what the surviving arms show about it is reported descriptively and is discussed in the deviations register, D-12 and D-18. This paper reports that outcome, what survived, and what it cost, with every deviation from the registered plan dated and listed. We are explicit about what this is not: it is not evidence that the original result is false. It is evidence that a protocol resting on temperature-0 determinism and on a strict output format does not transfer unchanged to open-weight judges served locally.

2 Related work

Judge models and their failure modes. Zheng et al. [3] introduced the MT-Bench and Chatbot Arena evaluations. Wang et al. [1], the study replicated here, isolates the swap-induced verdict reversal, quantifies it as a conflict rate, and proposes three calibration strategies, of which this study replicates one, Multiple Evidence Calibration. The 80-question item set we use is the one distributed in the original's own repository; we make no claim here about who first assembled it, because we did not verify that attribution.
Replication of machine-learning results. The target venue, ReScience C, publishes replications and accepts negative results; its author guidelines are the source for the venue requirements we follow.[4] We do not quote a definition of replication versus reproduction to a specific page, because we did not locate that wording on the guidelines page we fetched.
A concurrent pre-registration by the author,[5] registered on 2026-10-09, re-analyses five previously released judge-verdict datasets, none of them FairEval, for the gap between raw and chance-corrected agreement; it runs no judges, uses none of this study's verdicts, had run no analysis at registration, and the kappa-with-PABAK reporting used here was fixed in this study's own registration (sections 10 and 13), not imported from it.

3 Methods

The design was fixed in the pre-registration before data collection and is summarised here. Where this paper departs from it, the departure is numbered and dated in the deviations register.

3.1 Items and inputs

The items are the original authors' released question set, candidate answers and human labels, from https://github.com/i-Eval/FairEval.[6] That repository carries no licence file, so we redistribute none of its contents. Instead the inputs are fetched at run time from the original repository, pinned to commit `453c117a69159fd7b7de2f4e0519bac57bee21f4` and verified by SHA-256 before use. The verifier checks hash and byte length for all six files it pins and exits non-zero on any mismatch. Five are consumed by the arms that ran; the GPT-4 answer file is loaded and verified but never judged, because no arm uses it; it was tested green on a clean clone and tested to halt on a deliberately altered file.
The human labels are the original's own 80 annotations, with marginals ChatGPT 41, tie 14, Vicuna-13B 25. We considered re-annotating and rejected it: substituting new labels would replace the comparison standard and make our accuracy and kappa incomparable with the original's Table 5, which is the comparison the study exists to make.

3.2 Judges

Three open-weight instruction-tuned models from three families. The harness sends model TAGS to the local server, not digests; the digests below were read from the local model store after the runs and are recorded for identification, with the limitation set out in the deviations register, D-11: Qwen2.5-7B-Instruct (`845dbda0ea48`), Llama-3.1-8B-Instruct-q4_K_M (`46e0c10c039e`) and Mistral-7B-Instruct (`6577803aa9a0`). Those digests were read from the local model store, not written into the run artefacts as the registration required; see the deviations register, D-11. All inference ran locally on an Apple M4 Pro with 48 GB of unified memory. No paid API was used. A model that could not be pinned by digest would not have been used.

4.3 3.3 Prompts

Prompt provenance. The original prints two templates. Table 1 is the plain evaluation template, which asks for two scores on a single first line; we use it for the VANILLA arm. Table 3 is "the evidence calibration evaluation template that prompts LLMs to generate the evaluation evidence first, and then evaluate two responses", which is the MEC template. The `gen_prompt` in the released `FairEval.py`, lines 35-49 at the pinned commit, is a near-verbatim variant of Table 3: it carries a system message, labels the sections slightly differently, and asks for the explanation before the scores. We use that released variant for the MEC arms, and record it as a variant rather than as Table 3 itself.
The evidence that the released code is the MEC path is in the code. `query_gpt` hardcodes `temperature=1` and `n=args.k`, which is the paper's MEC setting of temperature 1 with k = 3; the `args.bpc == 1` branch performs the order swap; and the repository README invokes the script as "evaluation with MEC and BPC strategies". The released code cannot perform a temperature-0 vanilla run at all, because the temperature is fixed in the source.
The COMPARING prompt is a reconstruction and is declared as a deviation. The original does not print the template, in any figure or appendix or in the released code. It does specify the labels in prose, naming results " 'Assistant 1', 'Assistant 2', or 'Same' ", so the output space is given and only the prompt wording is ours.

4.4 3.4 Arms, seeds and controls

Every question in the VANILLA and MEC arms is judged in both presentation orders. Arms per judge: a Table 2 arm (VANILLA, temperature 0, both orders, two candidate pairs, SCORING template only, 320 calls); a Table 5 arm (MEC, temperature 1, k = 3, both orders, both templates, 960 calls); and an identical-order noise-control arm measuring each judge's self-inconsistency rate. That arm is narrower than the others: it runs in the AB order only and on the close candidate pair only (ChatGPT against Vicuna-13B), over both templates, 480 calls.
Departures from the original beyond the judge substitution, listed rather than left to inference:

- Seeds are fixed, which the original does not state: base 1234, MEC's k = 3 samples at 1234 + i, the noise arm at a disjoint block 1334 + i.
- All three judges are Q4_K_M quantised local builds, not full-precision weights. This applies to Qwen2.5 and Mistral as well as to Llama-3.1, whose tag names the quantisation explicitly.
- The COMPARING prompt is our reconstruction; the original prints it nowhere.
- The original's 100-run random-order average is computed by resampling our stored both-order verdicts rather than by 100 fresh inference passes.
- Verdict extraction uses a strict regex on the template's requested format, and outputs that do not match it are recorded as unparseable rather than coerced.
- MEC aggregation is a MAJORITY VOTE over the k = 3 parsed samples, with ties broken by first-seen order, and a record counts as parsed if ANY of its samples parsed. The original averages the numeric scores instead. This is a substantive difference in aggregation, not a detail.

Because a record counts as parsed if any one sample parsed, the record-level unparseable rate understates how often an individual generation failed. Section 4.3 therefore reports parse accounting at BOTH levels, per judge and per template.
Seeds, which the original does not state, were fixed in advance: base seed 1234, with MEC's k = 3 samples drawn at 1234 + i, because a single fixed seed at temperature 1 returns identical text and would silently collapse MEC to one sample. The noise-control arm uses a disjoint block, 1334 + i, so its verdict is an independent draw rather than a replay.

Five positive controls are implemented as `pc1_input_integrity`, `pc2_ground_truth_recovery`, `pc3_order_bookkeeping`, `pc4_determinism` and `pc5_parse_accounting`. PC-3 checks that each (question, pair, arm, template) cell holds exactly one AB and one BA verdict. It does NOT compare the stored candidate-answer hashes: those hashes (`r1_sha`, `r2_sha`) are recorded on every record but never read by the check, so PC-3 detects a missing or duplicated order, not a swapped candidate text. An earlier version of this paper described it as hash-based, which it is not. The first, second and fourth run before any arm; the third and fifth require verdicts and so run after the arms but before any hypothesis is examined. The point of all five is that a pipeline check could fire before any result existed: input integrity, exact recovery of the human-label marginals, order bookkeeping, temperature-0 determinism, and parse accounting in which unparseable verdicts are counted and reported rather than dropped.

4.5 3.5 Verdict extraction, and the determinism control as implemented

Two parsers exist. One is registered and produces every confirmatory number; the other is a secondary, exploratory parser declared after the data was seen. They are never mixed. THE REGISTERED PARSER. `parse_verdict` in `run_replication.py` has THREE extraction rules, one per template, and they are not the same rule. Stating them precisely matters, because the arm carrying every MEC result uses the most permissive of the three.

- SCORING, the paper's Table 1 template, used for the VANILLA arm. Requires the two numeric scores on the FIRST non-empty line, matched by the regex `SCORE_LINE` applied with `.match` to that line only. The pattern requires the whole line to be two numbers separated by whitespace, optionally with one decimal place each, and nothing else.
- SCORING_CODE, the authors' released variant, used for every MEC and noise arm. Applies `SCORE_LABELLED.search` to the WHOLE output, so the labelled score pair is accepted wherever it appears. This is deliberately more permissive than the Table 1 rule, because the authors' own prompt asks for the explanation before the scores.
- COMPARING, our reconstruction. Applies `COMPARE_LINE.match` to the first non-empty line and maps "Assistant 1", "Assistant 2" and "Same" to the three outcomes.

In all three cases the higher score, or the named assistant, becomes a win for that slot, equal scores or "Same" become a tie, and an output that does not match is recorded as unparseable with its raw text retained. Nothing is ever coerced into a verdict.
An earlier version of this section described only the first-line rule and attributed it to "the scoring templates". That was wrong for SCORING_CODE, and wrong about the arm that produces the MEC figures.
Because the surrounding harness has changed since registration for reasons unrelated to parsing, a whole-file hash is the wrong thing to pin. The two objects that must not change are pinned individually, and can be re-checked from the released code:

```
SCORE_LINE source text     sha256 38d9c3ca1b23d5a2
SCORE_LABELLED source text sha256 007446c7b728b8cd
COMPARE_LINE source text   sha256 10b434d6d06ebcbe
parse_verdict body         sha256 55af898e3576ab45
```

THE SECONDARY PARSER is `parse_verdict_relaxed` in `relaxed_parse.py`. It imports the unchanged `SCORE_LINE` from the harness rather than redefining it, so it cannot drift from the registered rule. It skips leading lines that consist only of a markdown heading or emphasis label, then applies the unchanged regex to the first remaining non-empty line, once. Its skip rule is broader than a markdown label: the pattern matches any line consisting only of letters and spaces, with an optional heading

marker, emphasis marks or trailing colon. A short prose line such as "Both responses are adequate" matches it and would be skipped. It does not skip bullets, numbered lines, or any line containing punctuation or digits. Its rule was chosen after the outputs were seen and the recovered count was known before it was written; both facts are recorded in the deviations register, D-5. Every figure it produces is labelled exploratory and is printed beside the registered figure, never instead of it.

THE DETERMINISM CONTROL, PC-4, AS IMPLEMENTED. Ten questions are drawn once with a fixed seed, `random.Random(20261010).sample(questions, 10)`, from the checksum-pinned question file, so the draw is identical for every judge and reproducible from the released code. The ten are 6, 23, 30, 33, 42, 43, 45, 46, 52 and 77. For each, the close pair is rendered through the paper's Table 1 template and submitted twice at temperature 0 with the same fixed seed 1234. The comparison is byte equality of the two returned strings; nothing is normalised first. A judge fails the control if any of the ten pairs differs.

THE HALT RULE. A PC-4 failure halts that judge's temperature-0 arm and is reported as a methods finding. It is not converted into a correction term, and it does not halt the judge's temperature-1 arms, which rest on the measured noise control rather than on PC-4. The failure is recorded in the results artefact as `pc4_ok` and `t0_arm`. One re-test under controlled conditions was pre-declared in the deviations register, D-1 before its outcome was known. It has run. Under that protocol Qwen2.5-7B-Instruct differed on 3 of 20 prompt pairs, a mismatch rate of 0.15 (`results/pc4_retest_declared.json`). It is exploratory and reinstates nothing.

4.6 3.6 Analysis

All statistical inference is paired on question and clustered on question, since the same 80 items recur in every condition. H1 and H3 were registered as exact McNemar per judge with Holm-Bonferroni correction across judges within family. H2 was registered differently, and we state it in the registration's own terms: "per judge, the signed difference between first-position and second-position win rates with a question-clustered bootstrap interval; then whether any two judges differ in sign with non-overlapping intervals". Interval types differ and are labelled wherever they appear. The H2 intervals and the c - s intervals use 10,000 question resamples. The random-order MEC accuracy brackets use 1,000 draws over presentation orders with the question set held fixed, which is a different resampling procedure answering a different question; the draw count is set in `analyze_mec.py`'s main analysis loop. Kappa is reported alongside PABAK in the results table, and the matched-marginal chance agreement pe and both full marginal distributions are printed for every cell by `analyze_mec.py`; they are not carried in the table itself. Kappa is never given as a bare number, because with ties at 17.5% of labels a kappa difference can be an artefact of the marginal distribution. The registration also asks for prevalence and bias indices;[7] those are defined for a two-category table and our verdict space has three categories, so they are not computed, and the deviations register, D-13 records that as an unmet registered requirement rather than substituting a look-alike number. marginal distribution.

Because the temperature-0 arm did not survive, no registered test of H1 or H3 was performed. H2 is also not evaluable as registered; what the surviving arms show about it is a post-registration descriptive deviation, set out in the deviations register, D-12 and D-18. Section 4 reports descriptive quantities from the arms that did survive, labelled as such.

5 4 Results

5.1 4.1 The headline: none of the three hypotheses is evaluable as registered

None of H1, H2 or H3 is evaluable as registered.
H1 reads from the temperature-0 VANILLA arm, and that arm did not survive for any of the three judges. H3 reads from it too, and was additionally unevaluable for reasons independent of the determinism failure, set out in the deviations register, D-9 and D-10: the VANILLA/COMPARING cell was never generated, and no counterpart to the original's calibrated quantity was ever computed.
H2 is also not evaluable as registered. Under a post-registration deviation its unchanged test was applied to the surviving MEC arms and reported descriptively; section 4.6 gives the figures and the limits on reading them.
An earlier draft of this paper reported H2 as evaluated and inconclusive. That classification was wrong, and it was corrected only after the null result was known; the deviations register, D-12 records the reclassification, its timing and the referee reading that prompted it. The H2 numbers changed when a later frame error was corrected; the deviations register, D-18 gives the detail.
Two judges failed the pre-registered determinism control, PC-4, which requires that re-running a prompt at temperature 0 with a fixed seed reproduce byte-identical output. The control checks ten prompts per judge. Qwen2.5-7B-Instruct failed on 1 of 10 (question 77), Mistral-7B-Instruct on 2 of 10 (questions 33 and 52), and Llama-3.1-8B-Instruct on 0 of 10. In both failing cases the registered rule halted the temperature-0 arm and the failure is reported here as a methods finding; it was not converted into a correction term, which the pre-registration explicitly forbids.
Llama-3.1-8B-Instruct passed PC-4 and ran all three arms. Its temperature-0 data is nonetheless unusable for the registered tests, for a different reason given in 4.3: under the registered parser only 1 of 160 question-pair cells has both presentation orders parsed, so there is effectively no paired data to test.
This is a negative result about the protocol under judge substitution, not a finding about positional bias. We did not measure positional bias well enough to say anything about it.

5.2 4.2 What the determinism failure means for the original's assumption

The original states that it sets "the generated temperature to 0 for deterministic generation results". On this stack that assumption did not hold for two of three judges. The effect is intermittent: the prompt that failed for Qwen reproduced byte-identical output on four subsequent isolated repeats. Those four repeats were themselves re-executions of the determinism check outside any declared protocol, before D-1 existed. They are not a second registered test and we do not treat them as one.
All three judges were checked on the same ten questions: the harness draws them with a fixed seed (`random.Random(20261010)`) from a question file pinned by commit and checksum, so the draw is identical across judges and reproducible from the released code. The three failures therefore fall on three different questions, with none shared between judges. That observation is weak on its own terms, 3 mismatches across 30 checks. A load or scheduling explanation would fit it; the D-1 re-test below tested that explanation and did not support it. So the observation is reported as a description of where the failures fell and not as evidence for any mechanism. Question 77 did recur for the same judge in that re-test, which is the one point of continuity across the two runs.
A single controlled re-test was pre-declared in the deviations register, D-1 before the outcome was known, and it has now run. It was exploratory and reinstated no halted arm. It does not support the load explanation. Under the declared protocol - serial requests, `OLLAMA_NUM_PARALLEL=1`, the judge model resident before the first call, nothing else issuing requests, all of which were INTENDED conditions rather than enforced ones,

since `pc4_retest.py` records the environment variable and verifies none of the four, so whether they held is unverifiable from the artefact -, Qwen2.5-7B-Instruct differed on 3 of 20 prompt pairs, a mismatch rate of 0.15: 1 of the 10 originally checked questions and 2 of 10 newly drawn ones (`results/pc4_retest_declared.json`, fields `n_prompts`, `n_mismatched`, `mismatch_rate`, `mismatched`). The differences are substantial rather than marginal, for example 332 against 307 output tokens on question 77.

So temperature-0 non-determinism persisted under the one controlled condition we tested, serial requests to a server we intended to be otherwise idle with the model resident, conditions the re-test script records but does not verify, which is narrower than "contention removed" in general. We do not compare the two rates: 3 of 20 against 1 of 10 cannot be distinguished at these sample sizes (Fisher exact, two-sided, $p = 1.00$), so the re-test establishes persistence and not a change in magnitude. We therefore do not attribute the non-determinism to load or scheduling. We also do not offer a replacement explanation: identifying the mechanism would need instrumentation below the inference server, which this study does not have. What we can say is that on this stack, for this judge, a fixed seed and temperature 0 did not produce reproducible output even when the server was handling one request at a time.

The practical reading, stated for the models we ran and not beyond them: on these locally served 7-8B models, temperature 0 did not guarantee identical output, and a conflict rate computed without a determinism check cannot separate run-to-run variation from positional bias. That is the ambiguity PC-4 was written to detect, and it detected it.

We make no claim about the original's own numbers. Its judges were different models served differently, and three locally served models on one stack cannot speak to them.

5.3 4.3 Parse accounting (PC-5), and one parser artefact

PC-5 requires unparseable outputs to be counted and reported per judge, never dropped. Counts for the arms that ran:

judge	arm / template	records	parsed (registered)	unparseable
Llama-3.1-8B	VANILLA / SCORING	320	7	313 (97.8%)
Llama-3.1-8B	MEC / SCORING_CODE	160	159	1 (0.6%)
Llama-3.1-8B	MEC / COMPARING	160	109	51 (31.9%)
Llama-3.1-8B	noise / SCORING_CODE	80	80	0
Llama-3.1-8B	noise / COMPARING	80	73	7 (8.8%)
Mistral-7B	MEC / SCORING_CODE	160	160	0
Mistral-7B	MEC / COMPARING	160	133	27 (16.9%)
Mistral-7B	noise / SCORING_CODE	80	80	0
Mistral-7B	noise / COMPARING	80	71	9 (11.2%)

A record counts as parsed if ANY of its $k = 3$ samples parsed, so the record-level rates above understate how often an individual generation failed. Sample-level accounting, which the record level hides:

judge	arm / template	records	unparsed records	samples	unparsed samples
Llama-3.1-8B	VANILLA / SCORING	320	313 (97.8%)	320	313 (97.8%)
Llama-3.1-8B	MEC / SCORING_CODE	160	1 (0.6%)	480	116 (24.2%)
Llama-3.1-8B	MEC / COMPARING	160	51 (31.9%)	480	359 (74.8%)

judge	arm / template	records	unparsed records	samples	unparsed samples
Llama-3.1-8B	noise / SCORING_CODE	80	0	240	69 (28.8%)
Llama-3.1-8B	noise / COMPARING	80	7 (8.8%)	240	159 (66.2%)
Mistral-7B	MEC / SCORING_CODE	160	0	480	9 (1.9%)
Mistral-7B	MEC / COMPARING	160	27 (16.9%)	480	250 (52.1%)
Mistral-7B	noise / SCORING_CODE	80	0	240	1 (0.4%)
Mistral-7B	noise / COMPARING	80	9 (11.2%)	240	116 (48.3%)
Qwen2.5-7B	MEC / SCORING_CODE	160	0	480	9 (1.9%)
Qwen2.5-7B	MEC / COMPARING	160	0	480	0
Qwen2.5-7B	noise / SCORING_CODE	80	0	240	0
Qwen2.5-7B	noise / COMPARING	80	0	240	0

Every arm that ran is listed, including Qwen2.5's four cells and the two noise cells an earlier draft omitted. Qwen2.5 has no VANILLA row because its temperature-0 arm was halted by PC-4 before producing any record, and Mistral-7B has none for the same reason. This changes how the MEC arms should be read. At record level Llama-3.1's MEC/SCORING_CODE arm looks almost clean at 0.6% unparseable; at generation level 24.2% of its samples failed to parse, and the majority vote over k = 3 concealed that. Llama-3.1's and Mistral-7B's COMPARING arms fail on half to three quarters of individual generations. Only Qwen2.5 is clean at both levels. Any statement that "the MEC arms are clean" holds at record level only, and we do not make it without that qualification.

The 97.8% figure needs explaining, because on its own it would imply that Llama-3.1 cannot score a pair of answers. It can. It emits a markdown label and then the two scores on the next line:

```
**Overall Scores:**
8.5 9.0

**Evaluation Explanation:**
```

The registered parser requires the two scores on the first line, so it does not reach them. This is a limitation of our measurement rule meeting a formatting habit of the model, and both halves of that sentence matter.

We did not change the registered parser. A secondary parser is declared in the deviations register, D-5, which records that its rule was chosen after the outputs were seen and that the recovered count was known before the rule was written. It skips leading lines that are only a markdown heading or emphasis label, then applies the unchanged regex to the first remaining non-empty line. It skips nothing else: not bullets, not prose.

Under that secondary parser, 281 of 320 VANILLA records parse, recovering 274. The 313 strict failures therefore split into two groups rather than one: 274 follow the markdown-

label pattern and are recovered, and the remaining 39 use other formats that the secondary rule does not reach and which we did not characterise further. Paired cells with both orders parsed go from 1 of 160 under the registered parser to 132 of 160 under the secondary one. Every figure from the secondary parser is exploratory in the sense of D-5 and is reported here beside its registered counterpart, never in place of it. And the recovery does not make this judge a clean one: Llama-3.1 remains above the registered 10% PC-5 flag under both parsers, at 97.8% unparseable under the registered rule and 12.2% under the secondary rule, so it is reported as a high-unparseable judge either way.

5.4 4.4 The COMPARING template carries its declared confound

COMPARING shows elevated unparseable rates on two of the three judges that ran it. Qwen2.5 parsed all 160 of its COMPARING records with none unparseable. The elevated rates belong to the other two judges: 31.9% and 8.8% for Llama-3.1, 16.9% and 11.2% for Mistral-7B. The same judges on the authors' own released SCORING prompt sit at 0.6% and 0.0%.

The pre-registration already declares our COMPARING prompt as a reconstruction, because the original prints it nowhere: not in the paper, which has no appendix, and not in the released code, which implements only the scoring prompt. The contrast above is consistent with the reconstruction being implicated in the parse failures, though Qwen2.5 parsed the same reconstruction perfectly, which points to an interaction between judge and template rather than to the template alone. We therefore report the COMPARING arm with that confound attached and draw no conclusion from it about judge behaviour.

5.5 4.5 What the surviving data supports

The MEC arms on the authors' own released prompt are clean AT RECORD LEVEL: 0.6% unparseable for Llama-3.1 and 0.0% for Mistral-7B. At sample level they are not, as 4.3 sets out: 24.2% of Llama-3.1's individual generations on that arm failed to parse and the majority vote hid it. Quantities internal to the MEC arms are therefore computable, including the self-inconsistency rate from the identical-order noise control. Those results are reported in 4.6.

What cannot be done is the comparison the study was built around. Every registered hypothesis contrasts something against VANILLA, and VANILLA is gone for two judges by the determinism control and effectively gone for the third by the parse rule.

5.6 4.6 Results from the surviving arms

Everything in this section is DESCRIPTIVE, including the H2 figures: they come from the registered test applied under a post-registration deviation to arms that are not the registered setting. Nothing here is a test of H1 or H3: both read from the temperature-0 VANILLA arm, which is halted for two judges and yields one contributing pair for the third. Computed by `analyze_mec.py` from `results/verdicts_*.json`.

Accuracy and Cohen's kappa are against the 80 released human labels, whose marginals are ChatGPT 41, tie 14, Vicuna-13B 25. Kappa is reported with PABAK, and the matched-marginal chance agreement pe is printed by `analyze_mec.py` for every cell though it is not carried in the table below, as the analysis plan requires.

judge	template	order	n	accuracy	kappa	PABAK
Llama-3.1-8B	SCORING_CODE	AB	80	0.588	0.315	0.175
Llama-3.1-8B	SCORING_CODE	BA	79	0.557	0.249	0.114
Mistral-7B	SCORING_CODE	AB	80	0.537	0.181	0.075
Mistral-7B	SCORING_CODE	BA	80	0.525	0.221	0.050
Qwen2.5-7B	SCORING_CODE	AB	80	0.600	0.259	0.200

judge	template	order	n	accuracy	kappa	PABAK
Qwen2.5-7B	SCORING_CODE	BA	80	0.512	0.226	0.025

Random-order MEC accuracy, computed by resampling the stored both-order verdicts as the pre-registration specifies in place of 100 fresh random-order runs. This is NOT the original's Balanced Position Calibration: BPC averages the numeric scores across both orders before deciding, and no BPC quantity was computed in this study (the deviations register, D-10). On SCORING_CODE the figures are, each with a 95% bracket from 1,000 draws over presentation orders, Llama-3.1 0.576 [0.519, 0.633], Mistral-7B 0.532 [0.463, 0.600] and Qwen2.5 0.556 [0.487, 0.625]. The original reports 53.2% for MEC on its SCORING template with ChatGPT as evaluator, so all three of our open-weight judges land around or slightly above it. An earlier version of this paper reported 0.429, 0.419 and 0.456 and concluded that all three sat below the original; that comparison was an artefact of the frame error recorded in the deviations register, D-16, and it reverses.
Conflict rate on the MEC arm, with the identical-order self-inconsistency rate from the noise control subtracted as section 4.2 of the pre-registration requires:

judge	template	n	c	s	c - s	95% CI
Llama-3.1-8B	SCORING_CODE	79	0.316	0.278	0.038	[-0.089, +0.165]
Mistral-7B	SCORING_CODE	80	0.375	0.237	0.138	[+0.000, +0.275]
Qwen2.5-7B	SCORING_CODE	80	0.475	0.225	0.250	[+0.100, +0.400]
Llama-3.1-8B	COMPARING	42	0.881	0.071	0.810	[+0.667, +0.929]
Mistral-7B	COMPARING	55	0.018	0.018	0.000	[-0.055, +0.055]
Qwen2.5-7B	COMPARING	80	0.287	0.075	0.212	[+0.100, +0.325]

Intervals are question-clustered bootstrap, 10,000 resamples, seed 1234, computed by `analyze_mec.py` function `cs_intervals`. n counts questions with both MEC orders parsed AND a parsed noise repeat, so it is smaller than the per-arm record counts; it is reported rather than left implicit. One of the three SCORING_CODE intervals, Qwen2.5's [+0.100, +0.400], strictly excludes zero. Mistral-7B's lower bound is exactly zero, so its interval does not exclude zero, and Llama-3.1's [-0.089, +0.165] includes it. Of the three COMPARING intervals, only Mistral-7B's [-0.055, +0.055] includes zero, while Llama-3.1's [+0.667, +0.929] and Qwen2.5's [+0.100, +0.325] exclude it. An earlier version of this paper said all three SCORING_CODE intervals excluded zero and that both COMPARING intervals included it; both statements were wrong, and the second miscounted the rows as well.
What c - s supports, and what it does not. The quantity c - s is the pre-registered operational measure of order dependence, so on the SCORING_CODE template we can say that these judges' verdicts change with presentation order beyond their own same-order noise by 0.038, 0.138 and 0.250 on the scoring template, with Llama-3.1's interval including zero. That is a descriptive statement about the MEC arm. It is not a test of H1, which is registered on the halted temperature-0 arm.
The accuracy gap between presentation orders is small on the scoring template once the frame error is corrected: 0.588 against 0.557 for Llama-3.1, 0.537 against 0.525 for Mistral-7B, 0.600 against 0.512 for Qwen2.5. An earlier version of this paper reported gaps of around 0.3 and built an argument about an unexplained order-dependent asymmetry

on them. Those gaps were an artefact of our own analysis error and that argument is withdrawn; the deviations register, D-16 gives the detail.
An earlier version of this paper said the H2 figures were unaffected by that correction because they were slot-frame quantities. THAT WAS FALSE and is withdrawn: two analysis functions each carried their own order-conditional remap, so the H2 table and the content-blind shares were wrong in every row. the deviations register, D-18 records it, and the figures below are the corrected ones.

Content-blind benchmark (exploratory, added after the results were seen; not registered) — A judge that ignored the answers entirely and chose by slot alone would still show an accuracy gap between the two orders, purely because the human labels are not symmetric across slots. This table asks how much of each cell's observed gap such a content-blind judge would reproduce, using that judge's own slot and tie shares.

judge	tem-plate	order	p(slot 1)	p(slot 2)	p(tie)	ob-served gap	content-blind gap
Llama-3.1-8B	SCOR-ING_CODE	AB	0.425	0.575	0.000	+0.031	-0.013
Llama-3.1-8B	SCOR-ING_CODE	BA	0.506	0.494	0.000		
Llama-3.1-8B	COM-PARING	AB	0.982	0.018	0.000	+0.121	+0.165
Llama-3.1-8B	COM-PARING	BA	0.925	0.075	0.000		
Mistral-7B	SCOR-ING_CODE	AB	0.613	0.388	0.000	+0.012	+0.045
Mistral-7B	SCOR-ING_CODE	BA	0.613	0.388	0.000		
Mistral-7B	COM-PARING	AB	0.015	0.000	0.985	+0.040	+0.026
Mistral-7B	COM-PARING	BA	0.015	0.000	0.985		
Qwen2.5-7B	SCOR-ING_CODE	AB	0.738	0.263	0.000	+0.088	+0.090
Qwen2.5-7B	SCOR-ING_CODE	BA	0.713	0.287	0.000		
Qwen2.5-7B	COM-PARING	AB	0.237	0.037	0.725	-0.025	-0.013
Qwen2.5-7B	COM-PARING	BA	0.163	0.225	0.613		

The content-blind prediction is computed on the SAME parsed questions as the observed figure beside it. An earlier version drew the human label marginals from all 80 registered questions while observing on parsed records only, so the two sides of each comparison rested on different question sets wherever a cell had unparsed records; three of the six cells changed when that was corrected. the deviations register records it.
Gaps are per template, not per order, so they appear once on each pair of rows. There is deliberately NO "fraction of the gap reproduced" column: after the frame correction several observed gaps are close to zero, Mistral-7B's on SCORING_CODE is 0.012, and a ratio of two near-zero numbers is numerically unstable and tells a reader nothing. Both gaps are printed in the same units, so a reader who wants the ratio can form it and will see how little it rests on.

The figure that matters most here is Mistral-7B on SCORING_CODE: p(slot 1) is 0.613 in BOTH orders. That is a stable preference for the first slot, invariant to which candidate occupies it, and it was invisible before the frame correction in the deviations register, D-18, because the earlier analysis re-flipped the BA records and made the two orders look like mirror images. Qwen2.5's content-blind gap on SCORING_CODE, +0.090 against an observed +0.088, is the one cell where slot shares alone account for essentially the whole observed gap.

This table is exploratory. It was not registered, it was added after the results were seen, and no mechanism is claimed from it.

These judges almost never emit a tie on the SCORING_CODE template, while the human labels tie on 14 of 80 questions. That caps achievable accuracy at roughly 0.825 before order enters, and it accounts for part of the distance between our numbers and the original's.

Verdict distribution per judge and template, over parsed verdicts, so a reader can see when a conflict rate is carrying information and when it is not:

judge	template	n	slot 1 ("A")	slot 2 ("B")	tie	
Llama-3.1-8B	SCORING_CODE	159	46.5%	53.5%	0.0%	
Llama-3.1-8B	COMPARING	109	95.4%	4.6%	0.0%	near-constant
Mistral-7B	SCORING_CODE	160	61.3%	38.8%	0.0%	
Mistral-7B	COMPARING	133	1.5%	0.0%	98.5%	near-constant
Qwen2.5-7B	SCORING_CODE	160	72.5%	27.5%	0.0%	
Qwen2.5-7B	COMPARING	160	20.0%	13.1%	66.9%	

Two cells are flagged near-constant. The flag marks any cell where one verdict category exceeds 90% of parsed verdicts, and that 90% threshold was chosen by us after seeing these data; it is a reporting flag and not a pre-registered rule, which is why the raw shares are printed above so a reader can apply any threshold they prefer. See the deviations register, D-7.

For the two flagged cells we still print the conflict rates. Only ONE of them should be read as uninformative, and the distinction is worth stating because it is not obvious: near-constant output drives the conflict rate in opposite directions depending on WHICH verdict is constant. Mistral-7B returns a tie on 98.5% of its parsed COMPARING items, and a constant tie cannot flip when the candidates swap, so its conflict rate c = 0.018 is near zero for an arithmetic reason and carries no information about order sensitivity. Llama-3.1 returns the FIRST SLOT on 95.4% of its parsed items, and a constant slot choice flips the candidate every time the order swaps, so its conflict rate c = 0.881 is near its maximum for the same arithmetic reason - and that is an informative position effect, not an artefact. The figures are those in the c - s table above, on the single common-set definition used throughout: Llama-3.1 COMPARING c = 0.881, s = 0.071, c - s = 0.810 on n = 42; Mistral-7B COMPARING c = 0.018, s = 0.018, c - s = 0.000 on n = 55. Earlier drafts carried a second pair of values for these cells computed over each quantity's own available set; the common-set definition is now used everywhere and its denominator is stated. Mistral-7B's net order-dependence c - s of 0.000, with a conflict rate c of 0.018, follows from returning a tie on 98.5% of parsed items. It is not evidence that the judge is unaffected by presentation order, and it must not be read that way.

The pre-registration has an interpretation rule for the opposite failure, a judge whose self-inconsistency approaches its conflict rate. It has no rule for a judge that answers almost identically on every item, and we have not invented one.

H2, as computed under the D-8 deviation (descriptive) — Signed first-position minus second-position win rate, per judge, on every surviving both-order arm. Intervals are at the Holm most-stringent level, 1 - 0.05/3 = 98.33%, for the reason given in the D-8 amendment. Computed by `analyze_mec.py` function `h2`, reproduced from the results files.

judge	arm	signed difference	98.33% interval	n records	excluded unparsed	questions dropped
Llama-3.1-8B	SCORING_CODE	-0.063	[-0.215, +0.089]	158	1	1
Mistral-7B	SCORING_CODE	+0.225	[+0.075, +0.375]	160	0	0
Qwen2.5-7B	SCORING_CODE	+0.450	[+0.312, +0.588]	160	0	0
Llama-3.1-8B	COMPARING	+0.884	[+0.767, +0.977]	86	51	23
Mistral-7B	COMPARING	+0.017	[+0.000, +0.069]	116	27	17
Qwen2.5-7B	COMPARING	+0.069	[-0.013, +0.156]	160	0	0

The registered decision rule asks whether any two judges differ in SIGN with NON-OVERLAPPING intervals. Every pair, on both arms:

arm	pair	opposite sign	non-overlapping	qualifies
SCORING_CODE	Llama-3.1 vs Mistral-7B	yes	no	no
SCORING_CODE	Llama-3.1 vs Qwen2.5	yes	**yes**	**yes**
SCORING_CODE	Mistral-7B vs Qwen2.5	no	no	no
COMPARING	Llama-3.1 vs Mistral-7B	no	yes	no
COMPARING	Llama-3.1 vs Qwen2.5	no	yes	no
COMPARING	Mistral-7B vs Qwen2.5	no	no	no

Each complete pair contributes two records, so n records is twice the number of contributing questions: 158 records is 79 questions for Llama-3.1 on SCORING_CODE, 86 is 43 questions for its COMPARING cell. Questions dropped are those where only one presentation order parsed, which the registration excludes from a paired test.

On SCORING_CODE one pair qualifies: Llama-3.1 leans toward the second slot (-0.063) and Qwen2.5 toward the first (+0.450), and their intervals do not overlap. That is the criterion H2's registered test names. On COMPARING no pair qualifies, because all three judges lean the same way.

WHAT THIS IS AND IS NOT. It is NOT a confirmation of H2. H2 is not evaluable as registered: the registered claim is scoped to the Table 1 temperature-0 arm, which produced no analysable data, and that is unchanged. What the table shows is a descriptive result on arms the registration did not name, computed under a post-registration deviation, on judges whose temperature-0 determinism we could not establish.

Two further reasons for caution, both of which weigh against us. First, an earlier version of this paper reported the opposite outcome on this arm, because the analysis was wrong; this result exists only because referees found that error, and it arrived in our favour, which is a reason for more scrutiny rather than less. Second, the surviving arms use Multiple Evidence Calibration, the original's own remedy for positional bias, so they are a weak setting in which to learn anything about a claim made on the uncalibrated template.

One reading in an earlier version also falls. Llama-3.1's COMPARING cell was described as "uninformative because the output is near-constant". Its corrected signed slot share is +0.884 with an interval of [+0.767, +0.977]. A judge that selects the first slot on almost every item is not uninformative about slot position; it carries the strongest slot-position signal in this study. The D-7's near-constant flag applies to the OUTPUT of both cells, but its uninformativeness conclusion holds only for Mistral-7B, whose constant verdict is a tie and whose conflict rate is therefore near zero by arithmetic. Llama-3.1's constant

verdict is a SLOT, so a swap flips the candidate every time and its conflict rate of 0.881 sits near its maximum for the same arithmetic reason. Both its conflict rate and its slot share of +0.884 measure one real position effect, and nothing about Llama-3.1's COMPARING cell is uninformative.

The pre-specified score-gap analysis (C-3) is not reported here. It needs the scoring template's numeric scores, and the registered binning was specified for the VANILLA arm.

5.7 4.7 One operational loss, disclosed

Qwen2.5-7B-Instruct's MEC arm was executed twice. The first execution completed and its output was then destroyed by an output-path collision in our own harness; the arm was re-executed and the re-execution is the data of record. The loss, what can and cannot be cross-checked, and the fix are recorded in the deviations register, D-6.

6 5 Discussion

What we can conclude. On two of these three locally served judges, Qwen2.5-7B and Mistral-7B, a temperature-0 setting did not guarantee identical output. Llama-3.1-8B PASSED the determinism control on all ten checked prompts, so the failure is not a property of every open-weight judge we tried, and we do not present it as one. Where it does occur, a conflict rate computed without a determinism check cannot separate run-to-run variation from order dependence. A pre-declared re-test under serial requests, run on the one judge that failed first, found the non-determinism persisted under the one controlled condition we tested, serial requests to a server we intended to be otherwise idle with the model resident, conditions the re-test script records but does not verify, which is narrower than "contention removed" in general, so we do not attribute it to load. We do not offer a replacement mechanism: identifying one would need instrumentation below the inference server, which this study does not have.

One limit on what the surviving arms could ever have shown about H2. They use Multiple Evidence Calibration, which is the original's own remedy for positional bias. A result on a calibrated arm is weak evidence about a claim the original made on the uncalibrated Table 1 template, whatever the interval around it, and that holds whichever way the result points. That is a reason to treat the H2 figures in 4.6 as descriptive, independent of their width.

What we cannot conclude. We make no claim about the original's numbers. Its judges were different models served differently, and three local models on one machine cannot speak to them. We also make no claim about positional bias as such. On the temperature-1 arms verdicts did change with presentation order beyond each judge's own same-order noise, though by less than an earlier version of this paper claimed: by 0.038, 0.138 and 0.250 on the scoring template, with one interval including zero, and most strongly by 0.810 on the reconstructed COMPARING prompt. We report those descriptively; but the registered test of position dependence was specified on an arm that did not survive, and we attach no mechanism to the accuracy asymmetry we observed, for reasons given in 4.6.

What a practitioner should take from it. If you intend to use an open-weight judge and rely on temperature 0 for reproducibility, verify that assumption on your own stack before you build on it, because it did not hold for two of our three judges. If you intend to parse a judge's output with a strict format rule, measure the compliance rate first: the registered parser could not reach the scores for 313 of one judge's 320 VANILLA records. Those 313 split into 274 that place the scores one line down behind a markdown label, and 39 in other formats we did not characterise. We did not assess whether any of those scores were substantively correct, only whether the registered rule could read them.

Limitations. Three judges on one machine. The COMPARING arm rests on a reconstructed prompt. Two COMPARING cells show near-constant output, and that cuts two ways: Mistral-7B's constant verdict is a tie, so its conflict rate is near zero by arithmetic and uninformative, while Llama-3.1's constant verdict is a slot, so its conflict rate is near its maximum for the same reason and measures a real position effect. One judge's MEC arm was executed twice after we destroyed the first execution's output through a filename collision in our own harness, and the two executions cannot be compared record by record. The MEC arm runs at temperature 1 with k = 3 samples per record, so two executions of it are not expected to agree verdict for verdict even on a perfectly deterministic stack. All of these are in the deviations register.
On contacting the original authors. We did not contact them. This replication therefore rests entirely on the published article and the released repository, and a reader should weigh it on that basis. We make no claim here about what the venue requires on this point.

7 6 Code and data availability

Our code and results are released in the project repository at https://github.com/dosvk/faireval-openweight-replication, archived at a Zenodo DOI obtained on acceptance, under the MIT licence required by section 9 of the registration. Released: the harness, the input verifier, the analysis scripts, the secondary parser, the determinism re-test, the stale-claim sweep and the build script; the per-judge verdict files with every individual sample and its raw text, with verbatim spans of the original benchmark's questions and answers redacted (see the data section of the registration, and the deviations register); the determinism re-test output; and the dated deviations register itself, as `DEVIATIONS.md` in the repository: it is released rather than reproduced in this article, so that every deviation stays recorded in full while the article keeps to the length of a ReScience C replication.
NOT released: the question set, candidate answers and human labels. Those come from the original authors' repository, which carries no licence file, so we redistribute none of them. The input verifier fetches them at a pinned commit and checks SHA-256 and byte length for all six files it pins before any run proceeds; five are consumed by the arms that ran. A consequence worth stating: if that upstream repository is deleted or rewritten, our artefact cannot be rehydrated from it, and the verifier will refuse to run rather than proceed on different bytes.
Also not released: the model weights, which are third-party. Inference ran locally through Ollama 0.30.8 on an Apple M4 Pro with 48 GB of unified memory. No paid API was used.
One reproducibility limitation, stated rather than left to be discovered. The registration committed to recording each arm's model digest in the results artefact before its run. That did not happen: the artefacts carry the model name and tag but no digest field. The digests appear in the deviations register and in section 3.2, but they were read from the local model store after the runs, so a reader cannot confirm from an artefact alone which weights produced it.
Expect to reproduce the pipeline and the controls. Do not expect byte-identical verdicts: section 4.2 reports that two of three judges failed a temperature-0 determinism check, so verdict-level agreement is a thing this paper measures rather than a precondition it relies on.

8 Deviations from the registered plan

Every departure from the pre-registered plan is recorded, with its date and its reasoning, in `DEVIATIONS.md` in the project repository released with this article (see Code and data availability). The register is the authoritative record: it holds each deviation in full,

including those that correct earlier entries of its own, with the original wording left intact beside every correction. It is kept in the repository rather than inlined here so that this article stays close to the length of a ReScience C replication while the record itself remains complete and auditable.

References

1. P. Wang et al. "Large Language Models are not Fair Evaluators." In: **Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)**. 2024, pp. 9440–9450. DOI: 10.18653/v1/2024.acl-long.511.
2. V. K. Sunkavalli. **Does positional bias in LLM-as-judge evaluation, and its calibration fix, hold for open-weight judges?** Pre-registration, OSF. 2026. DOI: 10.17605/OSF.IO/JW59C.
3. L. Zheng, W.-L. Chiang, Y. Sheng, and S. Zhuang. "Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena." In: **Advances in Neural Information Processing Systems** 36 (2023). DOI: 10.48550/arXiv.2306.05685.
4. ReScience C. **Author guidelines**. https://rescience.github.io/write/. accessed 2026-10-10. 2026.
5. V. K. Sunkavalli. **Chance-corrected agreement and multi-trial aggregation for LLM judges**. Pre-registration, OSF. https://osf.io/r93e2/. 2026.
6. i-Eval. **FairEval source repository**. commit 453c117a69159fd7b7de2f4e0519bac57bee21f4, accessed 2026-10-09. 2024. URL: https://github.com/i-Eval/FairEval.
7. T. Byrt, J. Bishop, and J. B. Carlin. "Bias, prevalence and kappa." In: **Journal of Clinical Epidemiology** 46.5 (1993), pp. 423–429.